Supplement Article

Integrative Assessment for Optimizing Aesthetic Outcomes When Treating Glabellar Lines With Botulinum Toxin Type A: An Appreciation of the Role of the Frontalis

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Abstract

Despite the perception that treatment of glabellar lines with botulinum toxin A is straightforward, the reality is that the glabellar region contains a number of interrelated muscles. To avoid adverse outcomes, practitioners need to appreciate how treatment of 1 facial muscle group influences the relative dominance of others. In particular, practitioners need to understand the independent role of the frontalis in eyebrow outcomes and the potential for negative outcomes if the lower frontalis is unintentionally weakened by botulinum toxin A treatment. In addition, practitioners must recognize how inter-individual variation in the depth, shape, and muscle fiber orientation among the upper facial muscles can affect outcomes. For optimal results, treatment of the glabellar complex requires a systematic and individualized approach based on anatomical principles of opposing muscle actions rather than a one-size-fits-all approach. This review provides the anatomical justification for the importance of an integrated assessment of the upper facial muscles and eyebrow position prior to glabellar treatment. In addition, a systematic and broad evaluation system is provided that can be employed by practitioners to more comprehensively assess the glabellar region in order to optimize outcomes and avoid negatively impacting resting brow position and dynamic brow movement.

Level of Evidence: 5

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Glabellar treatment with botulinum toxin type A (BoNTA) generally follows a standardized 5-point injection pattern intended to target the muscles underlying wrinkle formation. Although clinical trials utilize these standardized patterns for good reason, in the real world, patients' upper facial anatomies and desired clinical outcomes can significantly differ, necessitating a customized, rather than cookie-cutter, approach to treatment.[1] The routine assessment conducted before selection of a BoNTA injection pattern and dose is often solely focused on the type, extent, and strength of glabellar contraction. Because of an assumed risk of eyelid ptosis and inaccurate anatomical images in the literature, practitioners may be tempted to target glabellar lines by injecting BoNTA above the brow.[1–5] However, this approach often fails to target the true location of the underlying musculature that causes the wrinkles to form.[6–8] Further, because the upper face is an interconnected web of muscles, treatment of 1 muscle group influences the relative dominance of others and can affect the overall aesthetic outcome beyond the intended improvement in glabellar lines.[1,9]

Optimal outcomes following glabellar treatment include aesthetically pleasing changes in eyebrow position and shape as well as the elimination of glabellar lines.[10–13] Although preferences for eyebrow shape vary over time and between individual patients, excessive elevation of the lateral brow relative to the medial brow that results in an unnatural "quizzical" or "Spock" brow shape is typically considered to be aesthetically displeasing and unnatural.[11–14] Because the upper facial muscles have limited or no bony attachments, BoNTA treatment has the potential to negatively impact the delicate balance of forces that keep the eyebrows in position.[6,15] Hence, optimal outcomes can be reliably obtained only if the treating practitioner completes a thorough baseline assessment of the entire region, taking into consideration underlying alteration to the normal anatomy, the relative dominance of the various upper facial muscles, and eyebrow symmetry and position. Consensus recommendations for BoNTA treatment of the upper face are typically informed by clinical trials and the experience of expert clinicians, but this guidance is mainly focused on effacement of glabellar lines and rarely provide recommendations on how to balance the elimination of glabellar lines with optimization of brow outcomes.[6,9,16,17]

In this series of publications, we have assessed the effect of BoNTA for treatment of glabellar lines on eyebrow position and the impact of injection technique on brow outcomes and treatment duration.[18,19] In this review, we aim to provide an anatomical justification for the importance of an integrated assessment that includes the upper facial muscles and eyebrow position prior to glabellar treatment. We then propose a systematic and broader evaluation system that can be employed by practitioners to more comprehensively assess the region to optimize outcomes and avoid negatively impacting resting brow position and dynamic brow movement after treatment. The recommendations in this guidance are based on the current understanding of the functional anatomy of the upper facial muscles, recent data demonstrating the impact of glabellar treatment techniques on eyebrow position, and expert integrative analysis of glabellar technique with respect to functional anatomy.[6,19]

PRETREATMENT ASSESSMENT

Because the skin in the brow region is not anchored to bone, eyebrow position is determined by a balance between the opposing vertically oriented forces of the frontalis muscle and depressor complex (ie, procerus, depressor supercilii, orbicularis oculi, and corrugator supercilii).[6,15] Glabellar treatment with BoNTA can weaken the depressor complex and result in aesthetically pleasing eyebrow elevation.[18] However, if the suspensory activity of the lower frontalis is weakened by sufficient exposure to BoNTA, suboptimal brow outcomes may occur or underlying eyelid or eyebrow ptosis may be unmasked or worsened.[8,16,20–23]

An assessment for glabellar treatment (with or without forehead line treatment) should naturally focus on the type, location, and severity of glabellar lines and any areas of facial asymmetry in terms of eyebrow height and eye aperture.[24] Because the frontalis has a critical role in aesthetic outcomes following glabellar treatment, it should always be included in the assessment of patients who receive glabellar injections. Practitioners must consider the impact of individual differences in the anatomy of the brow and facial muscles, including the eyebrow depressors (procerus, depressor supercilii, orbicularis oculi, and corrugator supercilii) and the opposing eyebrow elevator (the frontalis), on brow outcomes.[8] These individual differences contribute to the complexity of identifying anatomical and functional landmarks and achieving consistently positive brow outcomes with glabellar line treatment.[6,16] A pretreatment checklist that incorporates assessment of the brow and facial muscles is outlined in Table 1.

Eyelid and Eyebrow Ptosis Assessment

Patients with underlying frank eyelid or eyebrow ptosis before glabellar line treatment are at risk of exacerbating those conditions or unmasking a compensated eyelid ptosis.[6,16,23,25] Although most events are transient and resolve spontaneously, eyelid and eyebrow ptosis are undesirable and, in some patients, can persist for several weeks or require treatment.[26,27] Hence, it is important to determine whether any preexisting overt or occult eyelid ptosis is present and to recognize the difference between true eyelid ptosis and pseudoptosis. Pseudoptosis is unrelated to

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Table 1. Pretreatment Checklist for Glabellar Line Treatment

Category	Descriptor
Medical history	1. Obtain medical history, including prior facial treatments or surgeries that may have altered patient's original facial anatomy 2. Review patient's goals and expectations
Assessment of facial anatomy and musculature	1. With frontalis muscle relaxed, assess for evidence of underlying subtle or occult eyelid ptosis and eyelid asymmetry[1,9,16] 2. Review eyebrow position to identify extent of any pre-existing asymmetry or eyebrow ptosis 3. Assess frontalis contraction pattern centrally and laterally and evaluate muscle position, mass, and symmetry under static conditions and dynamically during maximum frown and maximum brow lift[4,6,33,35] 4. Assess glabellar line contraction pattern by viewing glabellar lines at maximum contraction to identify wrinkle patterns and symmetry and relative contributions of muscular components of glabellar complex to glabellar line formation[24]
Photography	1. Take photographs of patient at rest, maximum frown, and maximum brow lift before treatment for comparison with outcomes at 2-week follow-up visit[1] 2. Photographs of eyes tightly shut will provide documentation of depressor activity of orbicularis oculi sphincter muscle 3. Discuss patient's anatomy and musculature with patient while looking at photographs, pointing out any unique features, asymmetry, or wrinkle patterns and highlighting any potential risk of adverse events; agree on expectations for outcomes

the muscles that directly contribute to eyelid position and may be caused by multiple factors, including excess skin, fat, or muscle in the eyelid area (dermatochalasis); eyebrow ptosis; or, in rare cases, trauma or neoplasms.[28]

Eyelid ptosis is common and, although it can be congenital, is usually caused by age-related dysfunction of the levator palpebrae superioris (levator), which is responsible for approximately 80% of eyelid elevation. Eyelid ptosis should be suspected in older patients as its prevalence increases with age, reportedly occurring in 2.5% to 11.6% of people aged 45 to 69 years and increasing to 33% to 43% of people older than 70 years.[28] Because underlying eyelid ptosis can be accentuated or unmasked with improper glabellar line treatment, it is important to understand the contribution of the frontalis muscle to eyelid elevation and ptosis, irrespective of patient age. As a secondary major contributor to eyelid elevation, the frontalis can raise the eyelid 1 to 2 mm above the upper margin of the iris at maximum lift (Figure 1A, B). Involuntary frontalis activity frequently compensates for underlying levator insufficiency and can be indicated by the presence of static forehead lines above the brow at rest (Figure 1C).[29] Therefore, weakening of the frontalis with BoNTA may unmask a compensated eyelid ptosis with no effect of the BoNTA on the levator muscle. To optimize the aesthetic outcome when the frontalis of patients of middle age and older is treated, this treatment should be co-administered with injection of the eyebrow depressor muscles; however, awareness of the potential impact on eyelid ptosis should be a top priority.

Because clinicians are generally trained to compare eye apertures on each side for asymmetries, compensated ptosis or mild bilateral ptosis can be missed on a cursory evaluation. When assessing patients for the presence of eyelid or eyebrow ptosis, the patient should be relaxed, upright, and looking straight ahead. A simple way to identify eyelid ptosis is to assess the amount of iris obscured by the upper lid; eyelid ptosis is considered to be present if the upper eyelid is positioned >2 mm below the upper margin of the iris/corneal limbus (Figure 2). In addition, any asymmetry between the eyes in the amount of upper eyelid that sits below the upper margin of the iris suggests that eyelid ptosis may be present. A more objective and precise measure of eyelid ptosis is to estimate the distance from the corneal light reflex to the upper eyelid margin, known as the marginal reflex distance (MRD) (Figure 2).[30] While the patient is looking straight ahead, a pen light is utilized to measure the distance from corneal light reflex on the mid-pupillary axis to the upper lid margin in each eye. For non-Asian individuals, an MRD of 3.5 mm to 5.0 mm is considered within the normal range, irrespective of sex, and an MRD <3 mm indicates that eyelid ptosis is present.[30,31] In general, the MRD for Asian individuals is 1.3 mm smaller than that observed in White individuals.[31]

Involuntary frontalis activity that may mask a mild or compensated eyelid ptosis may also be detected by examining the lower frontalis for horizontal forehead lines above the lateral aspect of the eyebrow. An indicator of this compensatory activity is the presence of static lateral forehead lines that may be unilateral or bilateral, depending on the pattern of the eyelid ptosis (Figure 2A). To definitively identify a compensated eyelid ptosis, one needs to first manually neutralize any involuntary frontalis activity. This can be done by (1) observing the lid position before and after gently tapping or stroking the skin above the eyebrow region, (2) gently fixing the skin of the forehead with the palm of the hand when the patient's eyes are closed and observing the lid position after patients gently re-open their eyes, or (3) by asking the patient to look down and then up without using their neck muscles while manually blocking their frontalis activity and observing the lid position before and during the time the patient takes to look up. If involuntary frontalis activity is present, then immobilization of the frontalis employing the techniques above will release the compensatory upward pull on the eyelid, and the medial and lateral brow position should drop. It may also be possible

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Figure 1. Role of the frontalis muscle in eyelid elevation. (A) In this 52-year-old female, the upper eyelid is located at the upper margin of the iris at rest. (B) At maximum lift, the frontalis muscle raises the eyelid 1 to 2 mm above the upper margin of the iris. (C) Lateral frontalis lines above the left eyebrow at rest suggest involuntary frontalis activity is compensating for levator weakness in this 47-year-old female.

to photograph the lid position before the involuntary frontalis activity returns, which often happens.

Eyebrow ptosis is also a common age-related condition that can be caused by the loss of supporting bone at the orbital margin, facial fat distribution, age-related changes in facial muscle activity, and soft-tissue elasticity.[32] The lateral region of the brow is particularly at risk of sagging with age because it has less support from the frontalis compared with the medial brow and, to some extent, because of the age-related loss of supportive fat volume over the lateral forehead and temples. Eyebrow ptosis can be detected by examining the relative height of the medial and lateral brow regions in both eyes at rest while the patient is looking straight ahead to identify overt areas of brow drop (Figure 3). Indicators of overt eyebrow ptosis include the natural hairy brow positioned below the orbital rim, a shortened brow-to-pupil distance, a disproportionate sag in the lateral brow, or brow asymmetry.[32]

Frontalis Muscle Assessment

Although individual differences in the frontalis muscle can contribute to differences in brow outcomes, the frontalis is often overlooked during facial assessments before glabellar line treatment.[17,19,33] As one of the few muscles with no bony attachments, the frontalis originates from the galea aponeurotica and inserts into the skin of the brow through a thin fascial layer.[6] The frontalis varies in depth, shape, and muscle fiber orientation across the forehead and may have the greatest degree of inter-individual variation among the upper facial muscles.[6,33,34] The position and mass of the frontalis can be assessed by viewing movement under maximum brow lift and by palpation—areas of stronger contraction can be identified by greater dynamic movement when patients lift their brows, deeper lines, and larger bulk during activity.[16] Although anatomical variation in the structure of the frontalis is the rule, for the purposes of evaluation before glabellar line treatment, individuals can be broadly dichotomized as having either a full sheet of muscle across the forehead or a frontalis that is divided centrally, to a greater or lesser degree, by a fibrous aponeurosis.[6,33,34]

In general, 2 functional states that describe the ability of the central frontalis to lift the medial eyebrows can be observed during active elevation of the brows. Both of these

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Figure 2. Eyelid ptosis severities. (A) No eyelid ptosis. In this 47-year-old female, the upper eyelids are located just slightly below the margin of the iris. However, lateral frontalis lines above the left eyebrow are suggestive of left eyelid ptosis masked by compensatory frontalis activity (marginal reflex distance = 4.5 mm for the upper right eyelid and 5 mm for the upper left eyelid). (B) Mild bilateral ptosis. In this 56-year-old female, the upper lid is located below the upper margin of the iris at rest, and approximately one-third of the upper iris is obscured (marginal reflex distance = 3 mm for the upper right eyelid and 4 mm for the upper left eyelid). (C) Moderate bilateral eyelid ptosis. In this 44-year-old man, the upper lid bisects the pupil in the right and left eyes at rest (marginal reflex distance = 0 mm for both upper eyelids), and 50% of the iris is obscured.

functional states have underlying anatomic correlates that are highly relevant to glabellar treatment. Patients with a full, uninterrupted frontalis muscle sheet across the entire forehead have the full force of central frontalis activity to oppose the downward forces of the depressor complex, evenly suspending the eyebrows.[6] These patients can be considered to have a "strong central pillar" of frontalis activity. Patients with a divided frontalis muscle, which is usually reflective of a central aponeurosis, may have little to no central frontalis activity.[6] Because these patients tend to have little to no muscle centrally, the bulk of the depressor complex has more limited or no direct antagonist muscle to oppose downward forces.[6,33,35] As such, these patients can be considered to have a "weaker central pillar"

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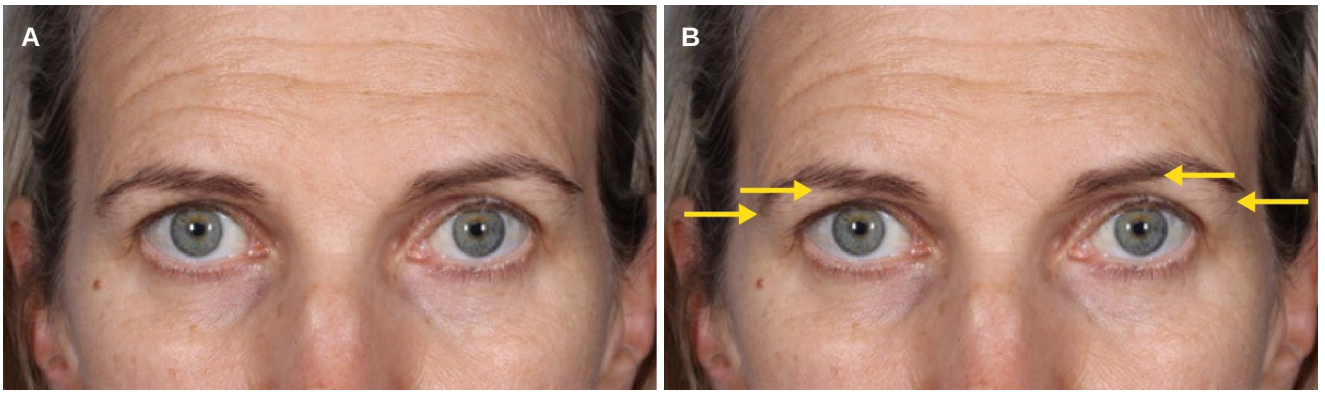

Figure 3. Eyebrow ptosis. (A) In this 52-year-old female, right eyebrow ptosis is present at rest and (B) is indicated by eyebrow asymmetry between the relative positions of the medial and lateral brows.

of frontalis activity and are considered to be at greater risk of an inadvertent drop in their medial brow and suboptimal outcomes following glabellar line treatment with standard injection patterns.[17,19,33]

Patients with a strong central pillar of frontalis activity can be identified by the presence of straight horizontal lines across the whole forehead on brow elevation, which, in some patients, may be present at rest (Figure 4A).[6,33,35] Straight horizontal lines are reported to be present in approximately 42% to 45% of people, and it is common upon eyebrow elevation in these individuals to see maintenance of a constant eyebrow arch with a similar degree of elevation of both the medial and lateral eyebrow.[33,35] Patients with a weaker central pillar of frontalis activity typically have wavy-shaped (Figure 4B) or mild to no (Figure 4C) central forehead lines upon maximum forehead lift.[6,33,35] Wavy-shaped or mild-to-no central forehead lines occur in approximately 45% to 58% of people and most likely reflect the presence of a midline frontalis aponeurosis.[33,35]

Glabellar Line Contraction Patterns

Although glabellar contraction lines vary from person to person, it is possible to identify a number of consistent glabellar contraction patterns that arise from variations in muscle strength and patterns of activity.[24,35] The relative contribution of each muscle responsible for glabellar contraction patterns can be identified in patients at maximum frown because lines tend to form at right angles to the pull of the contracting muscles (Figure 5A, B).[6,24] Vertical frown lines form primarily as a result of contraction of the corrugators, as the skin overlying the glabellar region moves medially, in the direction of the contraction.[35] "Hockey stick"–shaped lines, where the lower curved end of the "hockey stick" is positioned under the medial eyebrow, are thought to form as a result of depressor supercilii activity.[35] Horizontal line(s) at the root of the nose form with contraction of the procerus, which interdigitates with the lower frontalis, and short, vertical lines above the lateral brow are formed by the action of the superior (horizontally oriented) fibers of the orbicularis oculi and the lateral regions of the corrugators.[24] Deeper vertical lines above the lateral brow, in line with the midpupil, are thought to form at the site where the lateral regions of the corrugators attach to the dermis. In addition, the shape, depth, and position of vertical glabellar lines are influenced by glabellar soft-tissue thickness as well as a patient's skin thickness and skin type.[6]

IMPORTANCE OF FRONTALIS ASSESSMENT FOR OPTIMIZING AESTHETIC BROW OUTCOMES

As previously discussed, the most important factors determining medial eyebrow position are the balance of the caudal pull of the procerus and other depressors with the upward pull of the frontalis (Figure 6).[19] Practitioners need to be aware of the independent role of the medial and lateral regions of the frontalis in maintaining static brow shape and contributing to aesthetically pleasing dynamic brow elevation.[6] Patients with a weaker central pillar, with wavy lines indicative of significant central aponeurosis, are likely to be at a greater risk of a drop in the medial brow and therefore a suboptimal or unnatural brow outcome because of the limited muscle force available to suspend the brow from above. Therefore, in these patients, even a small amount of inadvertent exposure to BoNTA in the region where the procerus interdigitates with the lower frontalis (above the upper margin of the hairy brow) will weaken the frontalis muscle fibers that are present and result in medial brow drop (Figure 6).[17,19,20,33] This brow drop is usually visible at rest and can be exaggerated at maximum eyebrow lift.[19]

By way of illustration, representative images of patients with weaker central pillars (as characterized by the presence

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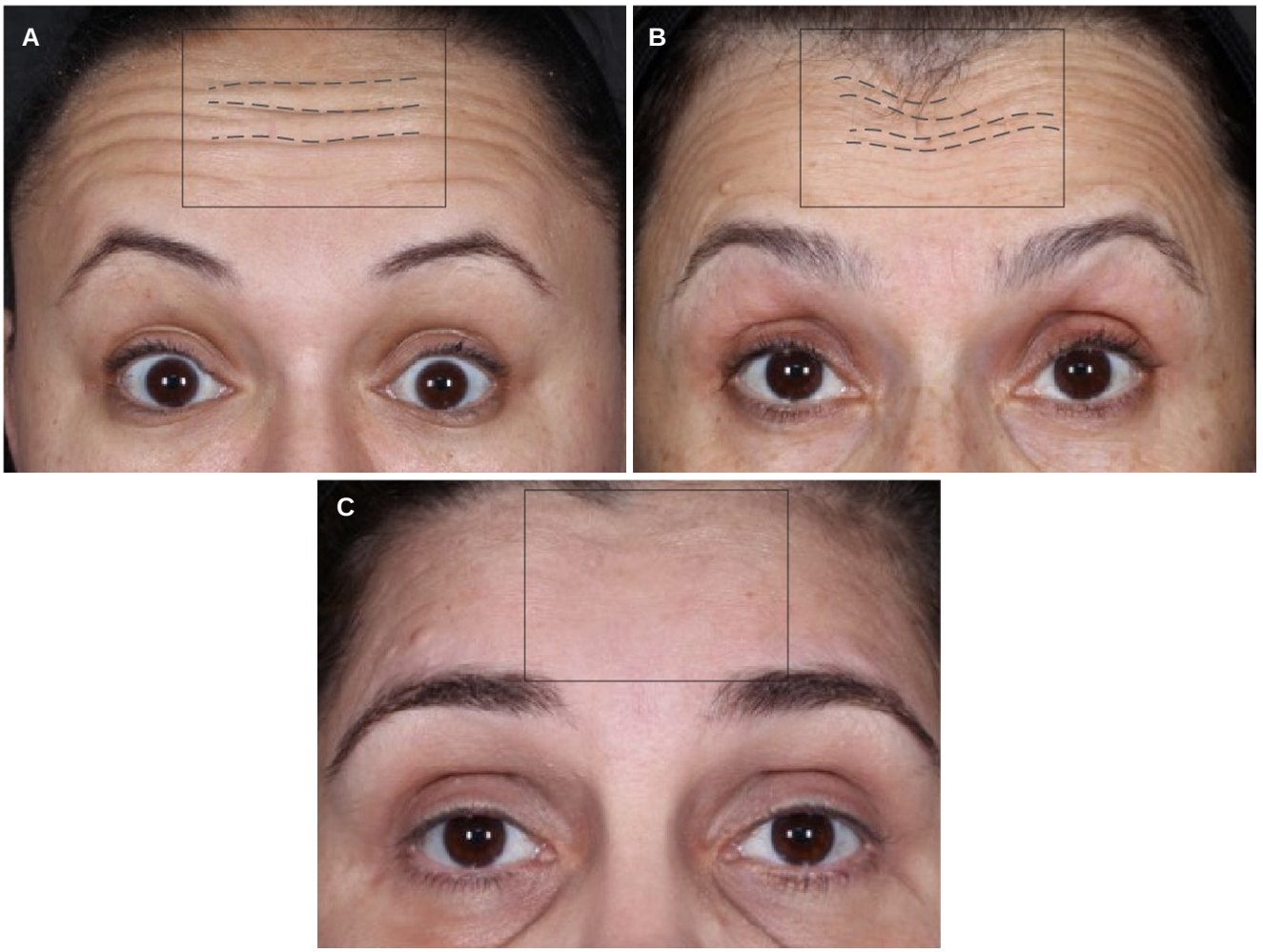

Figure 4. Forehead line patterns as indicators of frontalis muscle activity. (A) Straight central forehead lines at maximum lift are suggestive of a strong central pillar of frontalis muscle in this 41-year-old female. (B) Wavy-shaped central forehead lines at maximum lift are suggestive of a weaker central pillar of frontalis muscle in this 48-year-old female. (C) No central forehead lines at maximum lift are suggestive of a weaker central pillar of frontalis muscle in this 41-year-old female.

of wavy forehead lines) are shown in Figure 7. Both patients received glabellar treatment with DAXI 40 U administered as intramuscular injections employing a prescribed 5-point pattern comprising 1 injection to the procerus, 1 into the medial region of each corrugator, and 1 into the lateral region of each corrugator.[19] Patient A received a precise injection technique that more accurately targeted the origin of the corrugators and procerus, minimizing exposure of the frontalis to BoNTA, whereas patient B received a subtly different injection technique that inadvertently exposed the frontalis to BoNTA.[19] At 2 weeks following treatment, glabellar lines were eliminated, but there were differences in brow outcomes between the patients, which were visible at rest and were more evident at brow elevation. For Patient A, there were aesthetically pleasing brow outcomes and elevation of the eyebrows was balanced, with a mild lateral arch at rest and at maximum lift. In contrast, Patient B had suboptimal brow outcomes that were likely a result of a weaker central pillar, where there is limited frontalis muscle centrally, combined with inadvertent exposure of the central frontalis to BoNTA during injections in the lateral regions of the corrugators and the procerus. As such, the central frontalis was not able to oppose the downward pull of the depressor complex, and the patient was not able to lift their medial brow. Because the lateral frontalis was relatively unaffected, the patient was able to lift the lateral brow, which resulted in an unbalanced and unnatural dynamic appearance.

IMPORTANCE OF FRONTALIS FOR AVOIDING EYELID PTOSIS

A commonly held view is that eyelid ptosis can be avoided by placing the lateral corrugator injection points at least

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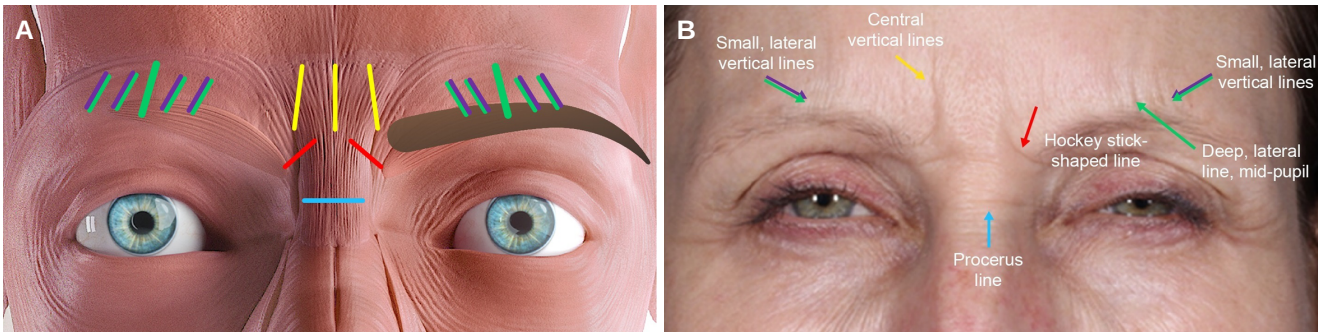

Figure 5. Muscle activity that contributes to glabellar line formation (A) is shown in a representative image of this 54-year-old female at maximum frown (B). The location of muscle activity and glabellar lines are shown by lines (A) and arrows (B), respectively, where blue denotes the procerus, yellow denotes the medial region of the corrugator, green denotes the lateral region of the corrugator, and purple denotes the orbicularis oculi.

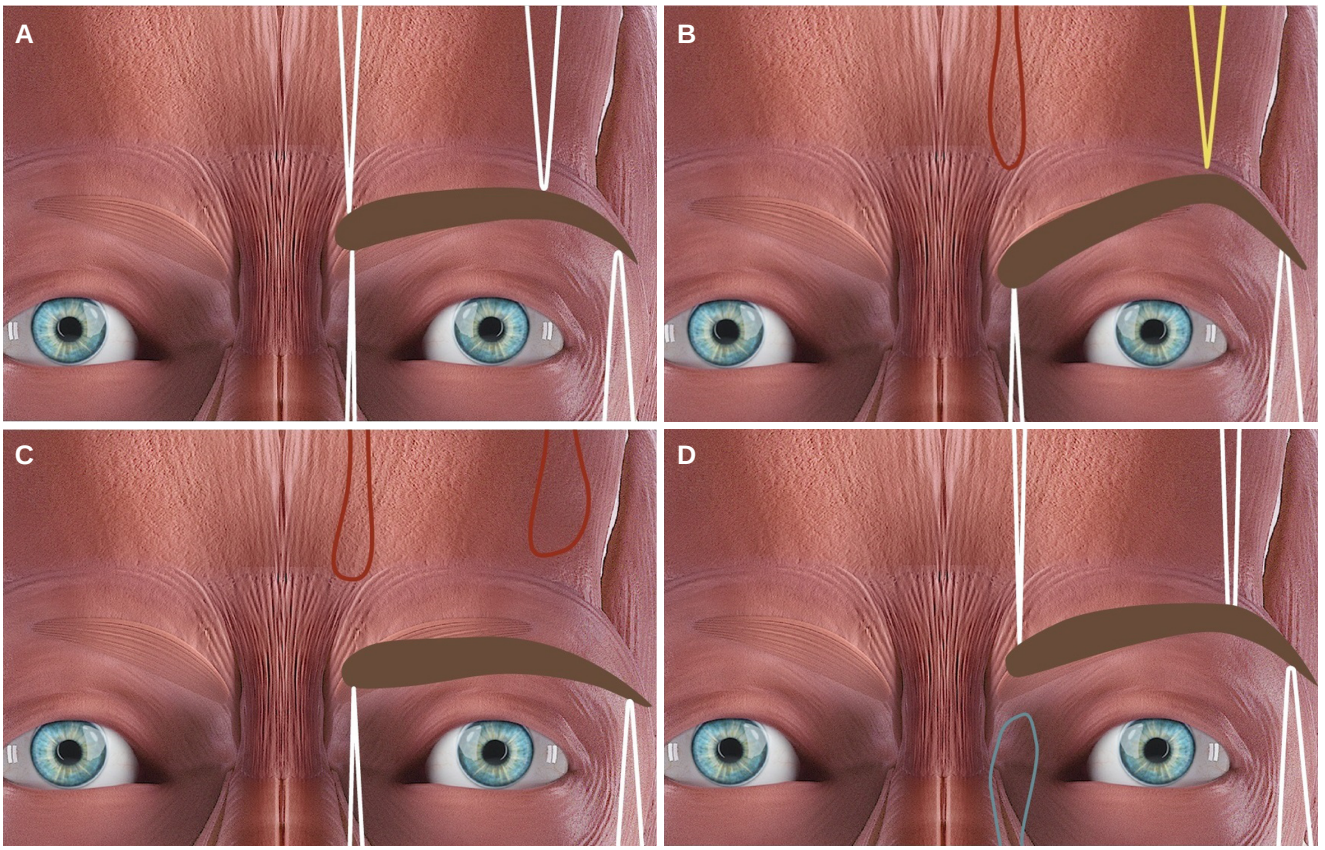

Figure 6. Balancing act of eyebrow position. (A) Eyebrow position is determined by a balance between the caudal pull of the depressor complex (procerus, depressor supercilii, orbicularis oculi, and corrugator supercilii) and the upward pull of the frontalis muscle. (B) Weakening of the suspensory activity of the lower frontalis (eg, by exposure to botulinum toxin) creates an imbalance in the forces holding the brow in position, resulting in suboptimal outcomes. This can occur through inactivation of the medial region of the lower frontalis, causing a drop in the medial region of the brow and elevation of the lateral region of the brow or (C) through inactivation of the medial and lateral regions of the lower frontalis, causing a drop across all regions of the brow. (D) Weakening of the downward pull of the depressor complex can elevate the medial region of the eyebrow, resulting in aesthetically pleasing outcomes.

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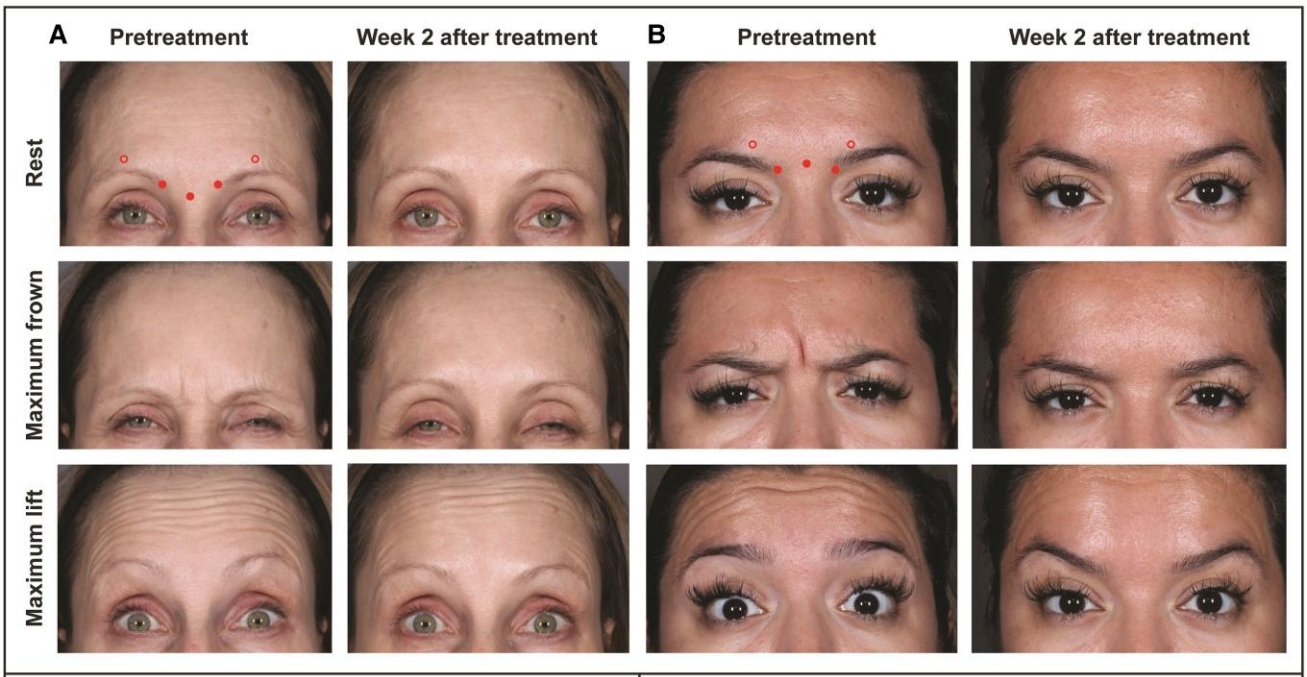

Pretreatment assessment • No indications of eyelid ptosis (MRD = ≥3.5 mm for both upper eyelids) • Eyebrow position symmetrical, with no evidence of eyebrow ptosis • Eyebrow elevation shows mild, central wavy-shaped forehead lines at maximum lift, suggestive of a weaker central pillar and a possible underlying aponeurosis • Lateral forehead lines present above both eyes at rest	***Pretreatment assessment*** • Indication of compensated right eyelid ptosis (ie, evident at rest only); the upper eyelid is below the upper margin of the iris, obscuring approximately one-half of the upper iris (MRD = 3 mm for the upper right eyelid and 4 mm for the upper left eyelid) • Evidence of eyebrow asymmetry at rest and at maximum lift; the lateral region of the right brow is slightly lower than the left brow • Frontalis contraction patterns showed no forehead lines in the lower region of the central frontalis at maximum lift and wavy lines in the upper forehead indicating a possible underlying aponeurosis
Pretreatment glabellar line assessment • Classic glabellar line pattern with central vertical lines and medial hockey stick–shaped lines above a mild horizontal crease, indicative of a balance between corrugator and procerus activity • Lateral glabellar lines above the pupil, which are particularly visible above the left lateral brow, indicate the site where the corrugator attaches to the dermis, with shorter lateral lines suggesting involvement of the orbicularis oculi and the lateral regions of the corrugators	***Pretreatment glabellar line assessment*** • A deep central vertical line without a visible horizontal procerus crease, indicating strong activity from the medial regions of the corrugators • Small vertical lateral glabellar lines, indicating activity from the lateral regions of the corrugators and the orbicularis oculi
Glabellar line treatment • Injections to the medial regions of the corrugators were placed deeply at the inferomedial aspect of the hairy margins of the brow, targeting the anatomical origin and region of most corrugator muscle bulk • Injections to the lateral regions of the corrugators were placed superficially between the mid-pupillary and lateral limbal lines, superior to the hairy brow • Injection to the procerus was placed deeply, in the midline and inferior to the horizontal upper margin of the medial brows	***Glabellar line treatment*** • Injections to the medial regions of the corrugators were placed deeply at the inferomedial aspect of the hairy margins of the brow, targeting the anatomical origin and region of most corrugator muscle bulk • Injections to the lateral regions of the corrugators were placed superficially, above the natural hair-bearing brow and between the medial canthal and medial limbal lines • Injection to the procerus was superior to the upper margin of the medial brows, too far above the procerus bony origin, and exposing the lower frontalis fibers to BoNTA
Treatment outcomes at 2 weeks • Glabellar lines were eliminated • There was balanced elevation of the brow, with a mild lateral arch at rest and at maximum lift • There were no discernible changes in the position of the upper eyelids, compared with baseline	***Treatment outcomes at 2 weeks*** • Glabellar lines were eliminated • The patient was no longer able to raise the upper eyelids above the corneal limbus at maximum lift compared to the pretreatment assessment • At rest, there were minimal changes in medial brow position and the lateral brows were arched • At maximum lift, the lack of medial brow movement was more evident and the lateral brows had an unbalanced, Spock-like appearance

Figure 7. Eyebrow outcomes following glabellar line treatment for patients with a weaker central frontalis in (A) this 54-year-old female with a lower central frontalis that was not exposed to botulinum toxin A (BoNTA) and (B) this 36-year-old female with a lower central frontalis inadvertently exposed to botulinum toxin A. In Patient B, injection points to the medial regions of the corrugators were accurately targeting the medial region, but injections to the lateral regions of the corrugators and the procerus exposed the lower frontalis to botulinum toxin A. MRD, marginal reflex distance.

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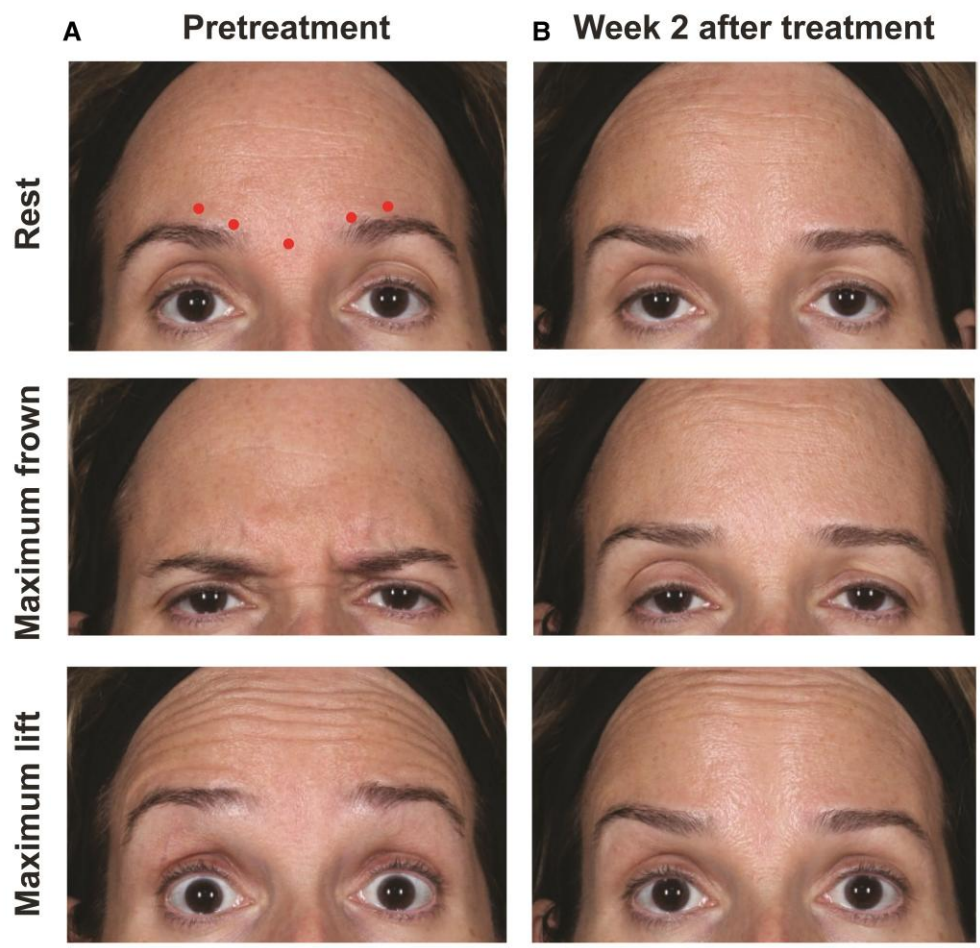

Pretreatment assessment

- Partially compensated mild bilateral eyelid ptosis evident, indicated bythe upper eyelid, which is positioned below the upper margin of the iris, obscuring approximately one-quarter of the upper iris and asymmetry between the eyes in the amount of upper lid showing (MRD = 3 mm for the upper right eyelid and 2 mm for the upper left eyelid)
- Evidence of eyebrow asymmetry. At rest, the right brow is lower than the left brow, which likely contributes to mild eyelid ptosis. In addition, a slight raising of the left lateral brow relative to the medial brow is suggestive of compensatory lateral frontalis activity in response to eyelid ptosis
- Frontalis contraction patterns show relatively straight central forehead lines, which are suggestive of a strong central pillar

Pretreatment glabellar line assessment

- Medial hockey stick–shaped lines with a visible horizontal crease below, indicating depressor and procerus activity
- Deep lateral glabellar lines above the pupil indicating the site where the corrugator attaches to the dermis
- Small vertical lateral glabellar lines indicating activity from the lateral regions of the corrugators and orbicularis oculi

Glabellar line treatment

- Injections to the medial regions of the corrugators were placed deeply, just above the hairy margins of the brow
- Injections to the lateral regions of the corrugators were placed deeply between the mid-pupillary and lateral limbal lines, superior to the hairy brow
- Injection to the procerus was placed deeply and slightly inferior to the horizontal line between the lower medial brow margins

Treatment outcomes at 2 weeks

- Glabellar lines were eliminated
- At rest, the distances between the mid-pupillary lines and the upper lid margins were slightly smaller in both eyes compared with baseline (MRD = 2 mm for the upper right eyelid and 1 mm for the upper left eyelid) and there was a drop in brow position across all regions, which was likely putting downward pressure on the upper eyelids

Figure 8. Aesthetic outcomes following glabellar line treatment for this 52-year-old female with a strong central frontalis and mild bilateral ptosis prior to treatment (A) and 2 weeks following treatment with botulinum toxin A (B).

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1.5 to 2 cm above the eyebrow, which should be distant enough to prevent BoNTA from diffusing into the orbital septum and weakening the levator.[5] However, there are rarely, if ever, any corrugator muscle fibers 1.5 to 2 cm above the central eyebrow. It has been repeatedly demonstrated from anatomical dissection and surgical experience that the bulk of the corrugator lies below the superciliary arch (supraorbital ridge) and, therefore, immediately deep to, or just inferior to, the hairy brow.[3,6,8] Therefore, in most patients, an injection >1.5 to 2 cm above the brow will directly target the inferior aspect of the frontalis muscle, which will at best drop the brow and at worst unmask a compensated eyelid ptosis.[19,23,36] For patients with a compensated eyelid ptosis, this strategy will almost certainly produce a frank ptosis, and, if the patient has a weaker central pillar, a concomitant drop in the medial brow will result. Thus, in patients in whom an underlying lid ptosis has been revealed or suspected and in those with a weaker central pillar, it is critically important to avoid weakening the lower frontalis. Therefore, in this population, the risk of a poor outcome will be reduced by precisely targeting the anatomical location of the medial regions of the corrugators and procerus and ensuring that the injections to the lateral regions of the corrugators are placed superficially, close to the brow.[6,16,26,36]

To demonstrate, representative images of a patient with underlying compensated eyelid ptosis who received BoNTA glabellar line treatment are shown in Figure 8. In this patient, glabellar complex treatment was administered as intramuscular injections employing a prescribed 5-point pattern comprising 1 injection to the procerus, 1 into each medial eyebrow region, intending to target the medial regions of the corrugators, and possibly placed above the medial brow in an attempt to avoid the levator of the eyelid, and 1 into the lateral regions of the corrugators. At 2 weeks following treatment, glabellar lines were eliminated, but aesthetic outcomes were not optimal. There was a drop in brow position across all regions, which was likely putting downward pressure on the upper eyelids. Although this patient had a strong central pillar, the drop in brow position was likely the result of (1) a frontalis that was exposed to BoNTA through corrugator injections that were placed too high and were, in fact, injected above the medial regions of the corrugators into the frontalis; as well as (2) high procerus injections, where the frontalis interdigitates with the procerus, thus weakening central frontalis activity; and (3) deep rather than superficial injections above the lateral brow, weakening frontalis muscle activity at this region. The right lateral brow was slightly elevated compared with the left lateral brow, resulting in asymmetry between the brows, and this asymmetry was more evident when viewed at maximum lift. Because the frontalis was overly weakened in this patient, the central frontalis was not able to maintain its compensation for the right lid ptosis that was present at baseline. As a result, only the lateral frontalis contracted, producing an unnatural cocked brow that was evident at rest, maximum frown, and maximum lift.

CONCLUSIONS

Although treatment of the glabellar complex may appear straightforward, a detailed understanding of facial functional anatomy is needed to anticipate the results of treatment and, therefore, achieve aesthetically pleasing results.[1,16] Before treatment, patients should be assessed for individual differences in facial anatomy that may affect the interactions between the frontalis and glabellar complex and the response of these muscles to treatment. Patients should also be carefully assessed for symmetry and underlying eyelid or eyebrow ptosis. In particular, when treating glabellar lines alone or in combination with forehead lines, practitioners should be aware of the central role of the lower frontalis in eyebrow outcomes and the potential for negative outcomes if the lower frontalis is unintentionally weakened by BoNTA treatment.[16,20,22,23] Avoiding direct injection of the lower frontalis will not only optimize brow position and minimize the risk of brow ptosis, but it can also reduce the risk of exacerbating any underlying eyelid ptosis, particularly in older patients with preexisting ptosis.[6,16,23,25] For optimal results, treatment of the glabella should be tailored based on a systematic and individualized approach underpinned by anatomical principles of opposing muscle actions rather than on a one-size-fits-all approach.

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